

COSIMA Cookbook: a computational learning module for analysing ocean and sea-ice model output

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Software

- [Review](#)
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Summary

The COSIMA Cookbook is an open computational learning module for analysing ocean and sea-ice model output in Jupyter notebooks. It has been developed by the Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean-sea ice model configuration (Kiss et al., 2020). The repository combines introductory tutorials with worked analysis examples, all exposed through a browsable Sphinx documentation site and backed by a citable archived release (COSIMA contributors, 2024).

The instructional structure is central to the project. “Cooking Lessons 101” introduces generic skills such as loading model output, working with labelled arrays, plotting, and interacting with shared data catalogues. These tutorials lead into domain-focused “recipes”: self-contained notebooks showing how to perform concrete diagnostics and analyses on ocean and sea-ice datasets. The current repository contains dozens of notebooks grouped into introductory, advanced, and region-specific collections, giving learners an incremental path from first contact with the data ecosystem to adaptation of full workflows for their own science questions.

The Cookbook grew from a practical need inside the COSIMA community: the tools used to analyse modern ocean model output are powerful, but the gap between package-level documentation and a reproducible end-to-end workflow remains large. Users need to understand not only Python and Jupyter (Kluyver et al., 2016), but also how to navigate high-dimensional model output, shared high-performance computing environments, and

43 domain-specific analysis conventions. The Cookbook addresses that gap by packaging
44 reusable workflows in the same medium in which users actually work.

45 Statement of Need

46 Ocean and climate model analysis has a steep entry cost. New users must learn how
47 to find datasets, load them efficiently, interpret metadata, operate on multi-dimensional
48 arrays, and produce scientifically meaningful diagnostics. General-purpose libraries such
49 as xarray (Hoyer & Hamman, 2017) provide the foundations for these tasks, but they
50 do not by themselves show a learner how to translate a research question into a robust
51 analysis workflow for a specific model and computing environment.

52 The COSIMA Cookbook fills that need with a domain-specific, openly maintained collection
53 of computational lessons and examples. Its contribution is not a new analysis package;
54 rather, it is a reusable learning resource that makes existing scientific software, data
55 conventions, and model output more approachable. This aligns well with JOSE's emphasis
56 on open educational resources that enable computational learning through authentic
57 practice.

58 Several design decisions make the Cookbook particularly useful for adoption by other groups.
59 First, each notebook is intended to be self-contained and well-documented, so learners
60 can read, run, and modify a complete workflow without reconstructing missing context
61 from scattered notes. Second, the repository distinguishes tutorials from recipes. Tutorials
62 teach transferable skills, while recipes demonstrate how those skills combine in realistic
63 analysis tasks. Third, the project includes contributor guidance and notebook review
64 conventions that encourage new analyses to be written in a reusable and pedagogically
65 clear style.

66 This structure is valuable beyond the immediate COSIMA context. Many research
67 communities maintain model output on shared infrastructure and face the same challenge:
68 turning expert tacit knowledge into examples that newcomers can adapt. The Cookbook
69 offers a model for how a scientific collaboration can capture that knowledge in version-
70 controlled notebooks, publish it as a living resource, and continuously improve it through
71 community contribution.

72 Educational Design and Experience of Use

73 The learning experience is organised as a progression. The documentation points new users
74 first to introductory lessons, including material on loading, slicing, and visualising model
75 output. Learners then move to more advanced tutorials and finally to recipe notebooks
76 that address concrete scientific questions. The categories of “appetisers”, “mains”, and
77 “local dishes” provide a lightweight pedagogical cue about expected complexity and scope.

78 The intended mode of use is also explicit. The repository is designed around Jupyter-
79 based analysis on the Australian National Computational Infrastructure, with guidance
80 on running notebooks through ARE/JupyterLab sessions and on accessing shared data
81 holdings. This makes the Cookbook more than a static collection of examples: it is
82 operational documentation for a real analysis environment. At the same time, the
83 notebooks demonstrate broadly transferable patterns for working with labelled geophysical
84 data, so individual lessons can be reused or adapted outside that environment.

85 The project also supports community authorship. Contributors are encouraged to submit
86 new notebooks through pull requests, document their methods clearly, and generalise
87 workflows so that they remain useful to others. In that sense, the Cookbook functions
88 both as a learning module for end users and as a framework for teaching reproducible
89 scientific communication through notebook design and peer review.

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